STS2005(06,07) **미적분학 I(HW1)**

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1. Use the $\varepsilon - \delta$ definition of a limit to prove:

$\boxed{2.4\text{-}\#18}$ $\lim_{x\to 1}(2x-5) = -3$

Proof. Now show that for any $\varepsilon > 0$, there is $\delta > 0$ such that for all $x \in \mathbb{R}$, if $0 < |x-1| < \delta$, then $|(2x-5)-(-3)| < \varepsilon$.
Let $\varepsilon > 0$ be given. Note that

$$\Big|(2x-5)-(-3)\Big| = \boxed{2}\ |x-1|.$$

Let $\delta = \boxed{\frac{1}{2}\varepsilon}$. Then $\delta > 0$ and for all $x \in \mathbb{R}$, if $0 < |x-1| < \delta$, then

$$\Big|(2x-5)-(-3)\Big| = \boxed{2}\ |x-1| < \boxed{2}\ \delta = \boxed{2\cdot(\tfrac{1}{2}\varepsilon)} = \varepsilon$$

$\boxed{2.4\text{-}\#32}$ $\lim_{x\to 2} x^3 = 8$

Proof. Now show that for any $\varepsilon > 0$, there is $\delta > 0$ such that for all $x \in \mathbb{R}$, if $0 < |x-2| < \delta$, then $|x^3 - 8| < \varepsilon$.
Let $\varepsilon > 0$ be given. Note that

$$\Big|x^3 - 8\Big| = (x^2+2x+4)\ |x-2|.$$

If $0 < |x-2| < 1$, then

$$1 < x < 3 \quad \text{and so} \quad (x^2+2x+4) < \boxed{19}, \quad (x^2+2x+4)\ |x-2| < \boxed{19}\ |x-2|.$$

Let $\delta = \min\{1, \boxed{\frac{\varepsilon}{19}}\}$. Then $\delta > 0$ and for all $x \in \mathbb{R}$, if $0 < |x-2| < \delta$, then

$$\begin{aligned}\Big|x^3 - 8\Big| = (x^2+2x+4)\ |x-2| &< \boxed{19}\ |x-2| \quad (\because |x-2| < \delta \leqslant 1)\\ &< \boxed{19}\ \delta \quad (\because |x-2| < \delta)\\ &\leqslant \boxed{19\cdot(\tfrac{\varepsilon}{19})} = \varepsilon.\end{aligned}$$

$\boxed{2.4\text{-}\#37}$ $\lim_{x\to a}\sqrt{x} = \sqrt{a}$ if $a > 0$.

Proof. Now show that for any $\varepsilon > 0$, there is $\delta > 0$ such that for all $x \geqslant 0$, if $0 < |x-a| < \delta$, then $|\sqrt{x} - \sqrt{a}| < \varepsilon$.
Let $\varepsilon > 0$ be given. Note that

$$\Big|\sqrt{x} - \sqrt{a}\Big| = \frac{1}{\boxed{\sqrt{x}+\sqrt{a}}}\ |x-a| \leqslant \frac{1}{\sqrt{a}}|x-a|.$$

Let $\delta = \boxed{\sqrt{a}\varepsilon}$. Then $\delta > 0$ and for all $x \geqslant 0$, if $0 < |x-a| < \delta$, then

$$\Big|\sqrt{x} - \sqrt{a}\Big| \leqslant \frac{1}{\sqrt{a}}|x-a| < \frac{1}{\sqrt{a}}\delta = \boxed{\frac{1}{\sqrt{a}}\cdot(\sqrt{a}\varepsilon)} = \varepsilon$$

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2.4−#30 $\lim_{x\to 2}(x^2+2x-7)=1$

Proof. Now show that for any $\varepsilon>0$, there is $\delta>0$ such that for all $x\in R$,
if $0<|x-2|<\delta$, then $|(x^2+2x-7)-1|<\varepsilon$

Let $\varepsilon>0$ be given. Note that

$$|(x^2+2x-7)-1|=|x+4||x-2|$$

If $0<|x-2|<1$, then

$$1<x<3 \text{ and so } |x+4|<7,\ |x+4||x-2|<7|x-2|$$

Let $\delta=\min\left\{1,\frac{\varepsilon}{7}\right\}$, Then $\delta>0$and for all $x\in R$, if $0<|x-2|<\delta$, then

$$\begin{aligned}|(x^2+2x-7)-1|=|x+4||x-2|&<7|x-2|\ (\because |x-2|<\delta\le 1)\\&<7\delta(\because |x-2|<\delta)\\&\le 7\cdot(\frac{\varepsilon}{7})=\varepsilon\end{aligned}$$

2.5−#47 $f(x)=\begin{cases}cx^2+2x & \text{if } x<2\\ x^3-cx & \text{if } x\ge 2\end{cases}$

Problem

For what value of the constant c is the function f continuous on $(-\infty,\infty)$?

Solution

$f(x)$ is continuous on $(-\infty,2)$ and also continuous on $(2,\infty)$

So, to prove that $f(x)$ is continuous on $(-\infty,\infty)$, check that $f(x)$'s limits on $x=2$

At x=2−, $\lim_{x\to 2-}f(x)=\lim_{x\to 2-}cx^2+2x=4c+4$

At x=2+, $\lim_{x\to 2+}f(x)=\lim_{x\to 2+}x^3-cx=8-2c$

At x=2, $f(2)=8-2c$

If 4c+4 = 8−2c, function f is continuous on $(-\infty,\infty)$

4c+4 = 8−2c => 6c=4 => $c=\frac{2}{3}$

Result

If $c=\frac{2}{3}$, fuction f is continuous on $(-\infty,\infty)$.

2.5−#54

Problem

Suppose f is continuous on [1,5] and the only solutions of the equation $f(x)$ = 6 are

$x=1$ and $x=4$, If $f(2)=8$, explain why $f(3)>6$ 20221585/유재혁

Solution.
If $f(3)<6$, we can apply one of the properties of the continuous function named Intermediate Value Theorem to this problem
check $f(x)$ on $[2,3]$, $f(2)>6$ and $f(3)<6$, there is $f(c)=6$ in $(2,3)$ but $f(x)=6$ is only possible at $x=1$ and $x=4$ so this is contradictory result.
So $f(3)\geq 6$
Suppose $f(3)=6$, but this is also a contradiction(which one is said on the above prove process)

Result
So $f(3)$ must be over 6 $(\because f(3)>6)$

2.5-#58 $\sin x = x^2 - x, (1,2)$

Problem
Use the Intermediate Value Theorem to show that there is a solution of the given equation in the specified interval

Solution
Let's say $f(x) = \sin x$, $g(x) = x^2 - x$

checking $g(x)$ in the interval $[1,2]$ $g(x)'s$ Min = 0 at $x=1$, $g(x)'s$ Max = 2 at $x=2$
checking $f(x)$ in the interval $[1,2]$, $f(x)'s$ Min is at $x=1$ and Max at $x=\frac{\pi}{2}$
$(\because 0<\sin 1<f(x)\leq 1)$
So calculating $f(x)-g(x)'s$ Max and Min in the interval $[1,2]$, $Max>0, Min<0$ is established. (because $\sin 1-0>0, \sin 2-2<0$)

Using the Intermediate Value Theorem, $f(x)-g(x)$ is continuous fuction, there is number c in $(1,2)$ such that $f(c)-g(c)=0$.

Result
there is number c in $(1,2)$ which makes $\sin x = x^2 - x$